Akash Ankush Kamble

kambleakash0@gmail.com +1(716)306-8297 /in/kambleakash /kambleakash0

Professional Summary

Accomplished Data Scientist and Engineer with hands-on experience at top-tier companies like Google (via acquisition) and OpenAI (via contract with Turing). Specializes in developing high-performance data pipelines and architecting scalable, end-to-end machine learning solutions. Proven track record of delivering significant performance improvements, such as optimizing a critical ETL pipeline by 24× and slashing regression testing cycles by 67%. Holds a Master's from IIT Madras and is currently completing a Master's in Data Science at SUNY Buffalo. Proficient in Python, Docker, GCP, and modern AI frameworks like LangChain and Llama Index.

Experience

Turing Inc. *(part-time)* **1 year, 5 months** — **Dec 2022 – May 2024**
Python Developer — *(Remote) Palo Alto, CA, USA*

- **RLHF Training for ChatGPT (Client: OpenAI)** | *Python, JavaScript, Ansible*
  - Authored and **curated** a high-quality **dataset of more than 800 novel programming conversations**, directly improving the model's reasoning and factual accuracy.
  - Consistently **maintained** a top-tier **quality score of 6.7/7 (over 95%)**, ensuring the effectiveness of the training data used for the next model iteration.
  - GPT4-o **yielded a 5× performance and speed improvement** over its predecessor, GPT4.

Argonaut AI LLC (acquired by Google) *(full-time)* **11 months** — **Dec 2022 – Oct 2023**
Lead Data Scientist — *(Remote) Berkeley, CA, USA*

- **Data Extraction Pipeline (Client: TeamCare)** | *Python, Docker, GCP*
  - Optimized a critical ETL pipeline, **achieving a 24× performance increase** by reducing average query processing time from 8 minutes to just 20 seconds.
  - **Engineered the backend service** to extract, clean, and generate vector embeddings from complex textual and tabular data for consumption by Large Language Models.
- **AI Studio (Client: Aspirion)** | *Python, Docker, Svelte*
  - Architected and **delivered a self-service "AI Studio"** that enabled non-technical teams to test and fine-tune LLM-based chatbots independently.
  - This **eliminated the need for engineering support** in the testing phase, significantly reducing development bottlenecks and accelerating the feedback loop.

HCLTech Ltd. *(full-time)* **3 years, 4 months** — **Aug 2019 – Nov 2022**
Senior Software Engineer — *(On-site) Chennai, TN, India*

Awarded *"O'Infinity Achiever"* trophy for top performance in automation projects.

- **CDM Platform Qualification (Client: Rubrik Inc.)** | *Python, Ansible, REST APIs, Redfish*
  - **Slashed** the regression **testing cycle time by 67% (from 3 weeks to 1 week)** by architecting and deploying a full-scale automation framework.
  - **Developed and packaged a custom Python library** that integrated with Ansible and hardware APIs with REST and Redfish, empowering the testing team with a streamlined, reusable solution.
- **AWS S3-like distributed object storage system (Client: Pure Storage)** | *Node.js, Docker*
  - **Engineered** a scalable, **S3-compatible distributed object storage system** for on-premises deployment, enabling the client to offer a robust private cloud storage solution.
  - Leveraged Docker Swarm for high-availability and load balancing, and **integrated the open-source Zenko CloudServer** with Pure Storage's proprietary backend infrastructure.

IIT Madras *(part-time)* **1 year, 11 months** — **Jul 2017 – May 2019**
Teaching Assistant — *(On-site) Chennai, TN, India*

- **Programming Instructor and Library Assistant (Client: Dept. of Computer Science)** | *C, C++*
  - Led instruction and **provided mentorship to freshman engineering students** on core programming principles through lectures, labs, and direct feedback on code assignments.
  - **Oversaw the management and circulation** of the Computer Science department's technical library, ensuring resource availability for the academic community.

Education

State University of New York (SUNY Buffalo) — **Jan 2025 – (expected) May 2026**
Master of Science in Engineering Sciences (Data Science) — *Buffalo, NY, USA*

Indian Institute of Technology Madras (IITM) — **Aug 2017 – Jul 2019**
Master of Technology in Computer Science and Engineering — *Chennai, TN, India*

Shri Guru Gobind Singhji Institute of Engg. & Technology (SGGSIE&T) — **Aug 2013 – Jul 2017**
Bachelor of Technology in Computer Science and Engineering — *Nanded, MH, India*

Technical Skills

Web Frameworks:	FastAPI, Django, Node.js
AI/LLM Tools:	LangChain, Llama Index, VertexAI, Gemini
DevOps Infrastructure:	AWS, GCP, Ansible, Docker, Hadoop, Jenkins
Languages:	Python, C, C++, R, HTML/CSS, JavaScript, SQL, Golang
Developer Tools:	Git, GitHub, Bash, Confluence, JIRA, LaTeX, Postman, VSCode, Codio, Redfish, REST APIs

Relevant Coursework

- Data Structures
- Software Engineering
- Algorithms Analysis
- Database Management
- Reinforcement Learning
- Cloud Computing
- Probability Theory
- Blockchain Foundation
- Distributed Systems
- Cryptography
- Operating Systems
- Digital System Design

Selected Projects

ML-Driven Protein Function Prediction | *Python, ProBERT, GCN* | **[Code]** — **Jan 2025 – May 2025**

- Developed a protein function prediction pipeline using a custom graph-aware k-NN model that achieved a **3x higher F1-score (0.45 vs. 0.15)** than a Graph Convolutional Network (GCN) baseline.
- The model accelerates research by enabling computational screening that can **reduce experimental validation workload by over 90%**. [Report]

MUBI's Movie Data Analysis & DB Pipeline | *Python, SQLite3* | **[Part 1, Part 2, 3]** **Feb 2025 – Apr 2025**

- **Executed an end-to-end analysis** of the MUBI dataset by building a normalized SQLite database from raw CSVs, **uncovering actionable insights** into MUBI subscribers' behavior and content trends about the hosted through targeted SQL querying and visualization. [Report]

Automated Content Summary Notifier | *Docker, FastAPI, Gemini, Postmark* | **[Code]** **Feb 2025 – Apr 2025**

- **Engineered a Python application to automate content consumption** by monitoring Substack blogs, **generating AI-powered summaries** with the Gemini API, and delivering them via Postmark email notifications. The service, built with FastAPI and containerized using Docker, provides users with timely, concise updates, significantly reducing reading time.

Dynamic Traffic Signal Control w/ Reinforcement Learning | *Python, SUMo* | **[Code]** **Jun 2018 – Apr 2019**

- Engineered a real-time, adaptive traffic signal control system using Reinforcement Learning to minimize congestion in a simulated urban environment (Eclipse SUMo).
- The dynamic agent demonstrated significant efficiency gains over static signal cycles, achieving:
  - A **6.5-hour reduction in total vehicle waiting time** for 8,000 vehicles.
  - A **66-minute improvement in clearing time** for 22,000 vehicles through a major intersection.
- This was my master's project at IIT Madras. [Thesis report]

Leadership | Co-curricular | Extracurricular

- **Volunteer**: Core team member for the event, IAD Days, a three-day annual AI symposium at the University at Buffalo, New York. — **Apr 2025**
- **Open-source contributor**: Llama-Index (Pull request) — **Sep 2023**
- **Open-source contributor**: Rubrik SDK for Python (Pull request) — **Sep 2022**
- **Assistant coordinator**: *M'trix* (an Ethical Hacking & Programming Club of SGGSIE&T) — **Sep 2013 – Aug 2015**
- **Seminar Conductor**: A 3-part talk series on Cloud Computing and Security (Slides) — **Oct 2015**
- **Filmmaker**: Shooting, editing and publication of a virtual campus tour of SGGSIE&T, Nanded (SGGSIE&T, A Voyage) — **Apr 2015 – May 2015**
- **Volunteer**: Core Team member for *RMageddon* (a robotics challenge at SGGSIE&T) — **Jan 2014**